

EMMA RAFKIN

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Education

Georgetown University (2023-2026)

M.S. in Computational Linguistics
GPA: 4.0

Dartmouth College (2017-2021)

B.A. in Computer Science. Minor in Linguistics
GPA: 3.74, Cum Laude

Employment

Johns Hopkins University Applied Physics

Laboratory (2021- Present)

Software Engineer/Data Scientist

- Created NLP tools that extract critical information from text for USG analysts.
- Awarded 6 grants to pursue research interests in summarization, entity disambiguation, adversarial AI, LLM domain adaptation, AI trustworthiness and multimodal anomaly detection.
- Led small research teams of 3-6 people, presented results to internal and external stakeholders
- 7 IP disclosures for original technology. 8 awards for leadership and innovation.

DALI Lab (2019 – 2021)

Software Engineer

- Produced full-stack web and mobile applications every 10 weeks.
- Created and ran classes on web design and development
- Sat on core executive board as a developer and Women in Science Program lead.

Alarm.com (2020)

Software Engineer Intern

- Created client-facing automated reports, to provide information to property managers about all devices in their units.

Cambly Inc.(2020)

Software Engineer Intern

- Enhanced Android application to include ESL courses for adults.

Dartmouth College CS Department (2018)

Teacher's Assistant

- Graded homework and test for OOP course, teaching a weekly review class to 10-15 students.

Camp Nicolet for Girls (2017-2018)

Counselor

- Organized and led daily activities for children aged 7-16 for 10 weeks.

Featured Research Projects

Cross-Lingual Transfer of Syntactic and Semantic Tasks: Use task vectors to evaluate the success of cross-lingual task transfer as language distance increases. (Master's Capstone Research Paper)

Task Arithmetic with Support Languages for Low-Resource ASR: Incorporate information from genetically related high resource languages to bolster performance on low resource ASR models. [\[System Paper\]](#)

Forget-Me-Not: Use task arithmetic to prevent catastrophic forgetting after LLM domain adaptation.

CaTE: Report on methods for data curation for trustworthy AI development. [\[Full Report\]](#) [\[Paper on domain knowledge elicitation\]](#)